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IU International University of Applied Sciences

Bachelor:
Computer Science

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Course name:
Project: Java and Web Development

Course code:
DLBCSPJWD01

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Date:
26.01.2025

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Abstract – Making Off the Adaptive Workout Plan

Idea and objectives of the project

The Adaptive Workout Plan was developed as a tool to provide easily accessible and free workout programs to individuals. According to the World Health Organization (WHO, 2024), obesity is a serious issue in the world, which must be addressed. Therefore, the goal was to produce a solution that is easily attainable and could help people who want to start exercising. Frameworks such as TypeScript, Tailwind CSS and Django were utilized to create a responsive single-page web application tailored to these goals.

Development process

The development process began with brainstorming and sketching how the solution could be designed. I quickly identified the need for a user-friendly and easily navigable single-page application accessible across all platforms. My primary objective was to minimize user clicks and make the interface as intuitive as possible. Once the visual design was concluded, I chose Django for the backend due to my familiarity with Python and its robust capabilities. For the database, SQLite was recommended by peers as an easy-to-use option for database beginners.

For the frontend, I selected TypeScript and Tailwind CSS, leveraging my prior experience with these technologies from earlier projects. Development involved extensive research using Google, Stack Overflow, and various generative AI tools. Additionally, I followed web app development courses on a platform called SoloLearn. While I aimed to simplify the app for both desktop and mobile users, I faced a steep learning curve when implementing backend functionality and integrating it with the frontend. Despite these challenges, the hands-on experience allowed me to learn significantly.

Explain design decisions

The design features three clickable boxes to minimize user effort and provide an easy entry point for individuals with low motivation, helping them overcome the famous doorstep mile that Scandinavians like to use as an excuse. The frontend is designed to be interactive and dynamic, with TypeScript enabling real-time updates based on the user's selected workout type and difficulty. For example, when a user picks 'Cardio' as the workout type, the difficulty levels and trainer message update instantly, making the process smooth and intuitive. My goal was to ensure the web application had a low threshold for engagement while delivering a functional and pleasant user experience. A responsive single-page application was chosen to maintain simplicity and accessibility.

To support users further, I included contact information for a personal trainer at the end of the workout selection, encouraging users to seek professional help if needed. A restart button was added to allow users to refine their workout plan without having to reload the page or exit the application. These decisions were informed by the overarching goal of creating a user-centric application that is approachable and supportive.

Highlight challenges

One of the biggest challenges was learning to implement backend logic and integrate a database. While I had some frontend experience, I was almost entirely new to backend development. Ensuring proper communication between the frontend and backend to deliver the correct solution required substantial effort. Debugging issues and understanding the relationship between Django, SQLite, and the frontend demanded a significant time investment.

Additionally, implementing responsive design across devices posed challenges, particularly in maintaining consistent styling and functionality. Tailwind CSS, while helpful for rapid development, required careful adaptation to meet design goals. This process often resulted in combining Tailwind CSS with the more familiar CSS to achieve the desired level of customization and polish.

Evaluation of the results

The project successfully achieved its goal of providing an easy-to-follow and intuitive web application that enables users to access workout plans with minimal effort. The solution is fully responsive, ensuring seamless functionality across desktop and mobile devices. The dynamic updates of the frontend, such as difficulty levels and trainer contact information, significantly enhance the user experience.

Moreover, pre-stored PDFs for workout plans made it easy for users to access the information they needed without delays, aligning well with the projects goal of creating a simple and accessible tool. The primary focus was to ensure that anyone could quickly and easily obtain a workout plan, reducing barriers to starting a fitness routine. The core features work as intended, providing a reliable and user-friendly solution.

Lessons learned

Reflecting on the process, I recognize that I was too eager to jump into development without fully understanding the frameworks I intended to use and how they worked together. In hindsight, dedicating more time to studying the technologies beforehand would have streamlined the process. However, I also believe that learning through failure is a valuable strategy in programming.

This project significantly improved my TypeScript skills and taught me that backend development and database integration are not as intimidating as they initially seemed. Additionally, I gained a better understanding of designing responsive applications using Tailwind CSS and ensuring that all components work cohesively. One key takeaway was the importance of debugging and testing in the stages to avoid compounding errors.

Finally, utilizing tools like Stack Overflow, various generative AI tools, and community forums, I was able to overcome roadblocks and better understand how to integrate different parts of the project. Additionally, I improved my skills throughout the entire design process, from brainstorming and creating UML diagrams to coding, debugging, and refining the project through feedback, ensuring that each phase contributed to the final product.

Reference list

World Health Organization. (2024, March 1). *Obesity and overweight*.
https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight